MATH 251 Course Project: Data Collection and Database Design - Cole Determan

Topic: IMDb Top Rated Movies & Directors

1. Dataset Description
This dataset contains a collection of 30 top-rated movies as listed on IMDb. It includes information such as the movie title, release year, genre, duration (in minutes), average user vote, and the director's name.

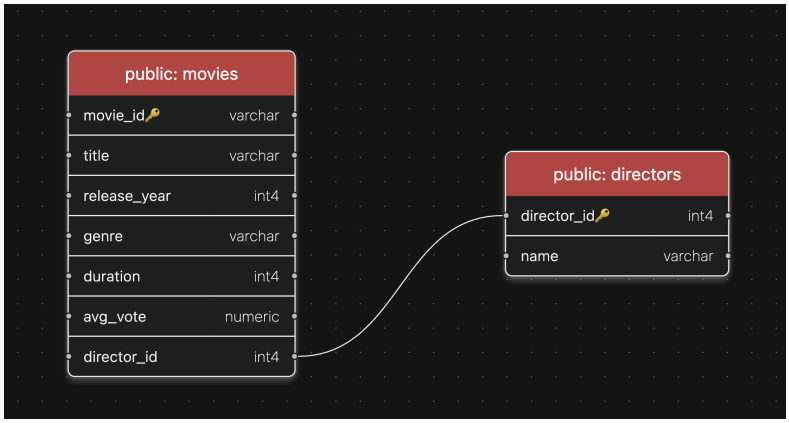
2. Data Source
The data was collected from the publicly available IMDb dataset via GitHub.
Original Source: [IMDb Dataset]
(https://gist.github.com/stungeye/a3af50385215b758637e73eaacac93a3)

3. Database Schema Design
I designed a relational database with two tables connected through a primary-foreign key relationship:

Table 1: directors
- director_id (Primary Key): A unique integer assigned to each director.
- name: The full name of the director (Unique).

Table 2: movies
- movie_id (Primary Key): The unique IMDb identifier (e.g., tt0111161).
- title: The name of the movie.
- release_year: The year the movie was released.
- genre: The primary genre of the movie.
- duration: Length of the movie in minutes.
- avg_vote: The average rating from user reviews.
- director_id (Foreign Key): Links each movie to its director in the directors table.

4. Relational Diagram (Conceptual)



movie_id	title	release_year	genre	duration	avg_vote	director_id
tt0120815	Saving Private Ryan	1998	Drama	169	8.6	1
tt0108052	Schindler's List	1993	Biography	195	8.9	1
tt0110912	Pulp Fiction	1994	Crime	154	8.9	2
tt0110357	The Lion King	1994	Animation	88	8.5	3
tt0120689	The Green Mile	1999	Crime	189	8.6	4
tt0111161	The Shawshank Redemption	1994	Drama	142	9.3	4
tt0038650	It's a Wonderful Life	1946	Drama	130	8.6	5
tt0102926	The Silence of the Lambs	1991	Crime	118	8.6	6
tt0482571	The Prestige	2006	Drama	130	8.5	7
tt0816692	Interstellar	2014	Adventure	169	8.6	7
tt1375666	Inception	2010	Action	148	8.8	7
tt0468569	The Dark Knight	2008	Action	152	9.0	7
tt2582802	Whiplash	2014	Drama	106	8.5	8
tt0110413	Leon: The Professional	1994	Action	110	8.5	9
tt6751668	Parasite	2019	Comedy	132	8.6	10
tt0099685	Goodfellas	1990	Biography	146	8.7	11
tt0114814	The Usual Suspects	1995	Crime	106	8.5	12
tt0133093	The Matrix	1999	Action	136	8.7	13
tt0114369	Se7en	1995	Crime	127	8.6	14
tt0137523	Fight Club	1999	Drama	139	8.8	14
tt0068646	The Godfather	1972	Crime	175	9.2	15
tt0043014	Sunset Blvd.	1950	Drama	110	8.4	16
tt0088763	Back to the Future	1985	Adventure	116	8.5	17
tt0109830	Forrest Gump	1994	Drama	142	8.8	17
tt0054215	Psycho	1960	Horror	109	8.5	18
tt0047478	Seven Samurai	1954	Action	207	8.6	19
tt0050083	12 Angry Men	1957	Drama	96	8.9	20
tt0073486	One Flew Over the Cuckoo's Nest	1975	Drama	133	8.7	21
tt0034583	Casablanca	1942	Drama	102	8.5	22
tt0167260	The Lord of the Rings: The Return of the King	2003	Action	201	8.9	23

(30 rows)

director_id	name
1	Steven Spielberg
2	Quentin Tarantino
3	Roger Allers
4	Frank Darabont
5	Frank Capra
6	Jonathan Demme
7	Christopher Nolan
8	Damien Chazelle
9	Luc Besson
10	Bong Joon Ho
11	Martin Scorsese
12	Bryan Singer
13	Lana Wachowski
14	David Fincher
15	Francis Ford Coppola
16	Billy Wilder
17	Robert Zemeckis
18	Alfred Hitchcock
19	Akira Kurosawa
20	Sidney Lumet
21	Milos Forman
22	Michael Curtiz
23	Peter Jackson

(23 rows)

5. Sample Questions Answered

1. Which movies in the top 30 were directed by Christopher Nolan?
2. What is the average user rating per genre?
3. Which directors have multiple movies in this top-rated list?
4. What is the longest movie in the collection?
5. How has the volume of top-rated films changed across decades?